Scand J Med Sci Sports 2011: 21: 765–772
doi: 10.1111/j.1600-0838.2010.01136.x

SCANDINAVIAN JOURNAL OF
MEDICINE & SCIENCE
IN SPORTS

Specific muscle adaptations in type II fibers after high-intensity interval training of well-trained runners

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Accepted for publication 8 March 2010

High-intensity interval training (HIIT) forms an important component of endurance athletes' training, but little is known on intramuscular metabolic and fiber type adaptations. This study investigated physiological and skeletal muscle adaptations in endurance runners subjected to 6 weeks HIIT. Eighteen well-trained endurance athletes were subjected to 6 weeks HIIT. Maximal and submaximal exercise tests and muscle biopsies were performed before and after training. Results indicated that peak treadmill speed (PTS) increased (21.0 ± 0.8 vs 22.1 ± 1.2 km/h, $P < 0.001$) and plasma lactate decreased at 64% and 80% PTS ($P < 0.05$) after HIIT. Cross-sectional area of type II fibers tended to have decreased ($P = 0.06$). No changes were observed in maximal oxygen consumption, muscle fiber type, capillary supply, citrate synthase and 3-hydroxyacetyl CoA dehydrogenase activities. Lactate dehydrogenase (LDH) activity increased in homogenate ($P < 0.05$) and type IIa fiber pools (9.3%, $P < 0.05$). The change in the latter correlated with an absolute interval training speed ($r = 0.65$; $P < 0.05$). In conclusion, HIIT in trained endurance runners causes no adaptations in muscle oxidative capacity but increased LDH activity, especially in type IIa fibers and in relation to absolute HIIT speed.

Muscle of endurance-trained athletes has a high oxidative capacity as well as an abundance of slow twitch oxidative (type I) fibers, with some fast twitch oxidative (type IIa) fibers, whereas resistance-trained individuals or athletes participating in high-intensity short intervals have a predominance of type IIa and fast glycolytic (type IIx) fibers (Gollnick et al., 1972). However, in more recent studies on well-trained competitive endurance runners, the proportion of type IIa fibers was higher than expected (~ 40%) and lactate dehydrogenase (LDH) activity correlated with fiber type (Kohn et al., 2007a, b). These findings are particularly important given the renewed interest in lactate as a central player in the integration of the metabolic response to exercise (Brooks, 2007).

High-intensity interval training (HIIT) forms an important part of the overall training regime of competitive endurance athletes (Billat, 2001; Esteve-Lanao et al., 2007). Recently, it was shown that male Kenyan runners training at higher speeds had a significantly better 10 km performance than Kenyan athletes training at lower speeds (Billat et al., 2003), despite the elite status in both groups. Depending on the type of sport (e.g. runners, cyclists or cross country skiers), this type of training may vary considerably in intensity relative to maximal oxygen consumption ($\dot{V}O_{2\,max}$), absolute interval speed, interval training duration, recovery between the intervals and the number of intervals (Billat, 2001). Even within one endurance discipline, differences in the above variables can influence the performance adaptations to HIIT (Kubukeli et al., 2002). However, it is clear that both performance and metabolic adaptations to high-intensity training interventions can occur within four to six sessions in either previously untrained or trained athletes (Weston et al., 1997; Kubukeli et al., 2002; Laursen et al., 2002).

There are only limited data available on muscular adaptations in well-trained endurance runners subjected to high-intensity training interventions (Kubukeli et al., 2002; Midgley et al., 2007). When controlled HIIT protocols were used to study the effects on muscle adaptations in various types of endurance athletes, the results were not consistent (Houston & Thomson, 1977; Kraemer et al., 1995; Weston et al., 1997). This may be partly because the prior training status of the athletes may preclude further adaptations in the parameters investigated. Alternatively, the intervals were not sufficiently individualized for the duration, as would be the case when elite athletes train (Billat, 2001).

Improvements in performance as a result of HIIT are well established (Billat, 2001; Kubukeli et al., 2002), but significant improvements in LDH enzyme activities in response to HIIT are only observed in previously untrained or moderately trained individuals (Linossier et al., 1997). We reported previously that in well-trained endurance runners, LDH activity correlated with fiber type (Kohn et al., 2007a), but fiber type explained only 36% of the variance in LDH activity.

Aim of the study

The aim was to investigate the influence of 6 weeks of HIIT in well-trained endurance runners on physiological markers of performance and with particular attention to adaptations in muscle oxidative and glycolytic capacity. In accordance with the intensity of training requiring speeds that participants did not usually include in their training, the hypothesis was that the majority of change in enzyme activities and muscle fiber morphology would occur in fast twitch fibers.

Methods

Subject recruitment and training volume assessment

The Ethics Committee (Sub-Committee C of Research Administration) of Stellenbosch University approved the study. Eighteen endurance runners (three colored, seven Xhosas and eight Caucasians), racing competitively between 3000 m and the half marathon, were recruited. Inclusion criteria were: (i) actively competing in races, (ii) a current 10 km personal best (PB) <38 min and (iii) a clean bill of health for the past 6 months. Each signed a written informed consent before participation and completed a detailed questionnaire, recorded race distances and times, training type, total distance and training duration, which was confirmed by their respective coaches (Kohn et al., 2007a). Favorite cross country, track and road distances were reported, from which an average preferred racing distance for each was calculated. The group competed in a 10 km field test to validate the current 10 km performance status (Table 1b). Because of a staggered entry into the actual laboratory-training phase, no 10 km field test was performed after HIIT. Ten of these athletes were screened 3 months before the intervention to assess performance, training and maximum exercise variability before the HIIT intervention (Table 1a). This pre-screening phase did not involve submaximal exercise tests, blood sampling or the taking of muscle biopsies, and primarily served to monitor training consistency before the intervention, and the change in training during and after the intervention.

Training volume before and after the HIIT intervention consisted of outdoor and laboratory training, including laboratory tests. "Before HIIT" training consisted of data recorded 2 weeks before HIIT, whereas "after HIIT" training was performed during the 6 weeks HIIT and 2 weeks post-HIIT (expressed as km/week).

Exercise tests and muscle biopsies

Athletes were familiarized with treadmill running before any testing was performed. Tests were performed on separate days (at least 2 days between tests) to allow complete recovery. The group was encouraged to be well rested and abstain from races before the testing day. All tests and procedures listed below were performed before and after the intervention.

$\dot{V}O_{2max}$ testing and peak treadmill speed (PTS)

Each subject completed two maximal exercise tests on a treadmill (RunRace, TechnoGym) as described previously (Kohn et al., 2007a). Briefly, after a 5-min warm-up and rest, athletes started running at 14 km/h for 30 s, after which the speed was increased by 0.5 km/h every 30 s until voluntary exhaustion. Heart rate (HR), $\dot{V}O_2$ (mL/min/kg), respiratory exchange ratio (RER) and minute ventilation ($\dot{V}_E$) (Jaeger Oxycon Pro, Hoechberg, Germany) were recorded throughout the test. PTS was calculated as follows, taking each second into account:

$$\text{PTS} = \text{Completed full intensity (km/h)} + \frac{\text{(seconds at final intensity)}}{30\,\text{s}} \times 0.5\,\text{km/h}$$

Submaximal exercise test and blood sampling

Submaximal exercise tests were conducted as described previously (Kohn et al., 2007a), with workloads (5 min each with 1-min rest between workloads) corresponding to 64%, 72% and 80% of each individual before HIIT PTS. Directly after each workload, a blood sample was collected from an intravenous catheter for plasma lactate determination. Recording of cardiorespiratory parameters was performed as for the maximum exercise tests. The same absolute workloads were performed after the intervention.

Muscle biopsies

Biopsy samples were taken from the *Vastus lateralis* muscle as described (Kohn et al., 2007a). This was carried out 5–7 days before the 2 weeks of baseline testing and directly after the post-HIIT exercise testing. All subjects had biopsies, although not all analyses could be performed on all samples due to sample size. The sample was frozen in liquid nitrogen and stored at -87 °C until analyses.

HIIT protocol

The HIIT protocol was based on that described by Smith et al. (2003), with modifications. HIIT sessions were performed twice a week for 6 weeks on a treadmill. All sessions were supervised. Individual interval training speed (ITS) was calculated as 94% of the PTS attained from the maximal exercise test. Interval duration was determined from a separate test. Each athlete ran at ITS until exhaustion (T_{max}) and the interval duration was calculated as 60% T_{max} (averaged 2.7 ± 0.5 min). Recovery between intervals was calculated as half of the interval duration. Therefore, the HIIT protocol was defined as

$$6\times\ \text{intervals at 94\% PTS for } 60\%\,T_{max}$$
$$\text{with } 1/2(60\%\,T_{max}) \text{ as recovery}$$

Biochemical analyses

Plasma lactate concentrations

Plasma lactate concentrations (mmol/L) were determined using a commercially available kit (Lactate PAP, bioMérieux, Marcy l'Etoile, France) according to the manufacturer's instructions.

Muscle morphology

Cross-sectional area (CSA, μm^2) of types I and II muscle fibers was determined using before and after samples from 10 athletes using methods described previously (Kohn et al., 2007a). The size of the biopsy samples obtained excluded eight subjects from this analysis. The average number of capillaries surrounding a fiber was determined histologically using the amylase PAS stain (Andersen & Henriksson, 1977) without taking into account fiber type or size (capillaries/fiber).

Myosin heavy chain (MHC) content and enzyme activities in homogenate muscle samples

MHC isoform content was separated on SDS-PAGE gels, scanned and the band intensities were quantified using a software package (CREAM 1D, KEM-EN-TEC, Taastrup, Denmark) (Kohn et al., 2007b). Isoforms corresponding to MHC I, MHC IIa and MHC IIx were expressed as a percentage of the total.

Activities of citrate synthase (CS), 3-hydroxyacetyl coenzyme A dehydrogenase (3-HAD) and LDH in homogenate muscle samples were determined fluorometrically and expressed as μmol/min/g dry weight (dw) (Kohn et al., 2007a).

Single fiber identification and LDH activity in pools of typed fibers

From each of the 14 athletes, ~ 150 single muscle fibers were dissected from freeze-dried samples, classified into fiber types using SDS-PAGE and pooled. Fibers were classified as either pure (expressing only MHC I, IIa or IIx) or hybrids (I/IIa or IIa/IIx). Pooled fibers from two subjects were unfortunately lost during the weighing process. LDH activities of types I and IIa fiber pools were determined for 12 pairs of subjects (before and after HIIT) according to Kohn et al. (2007a). Briefly, pooled fibers were weighed using a Cahn 25 balance (Thermo Fischer Scientific, Waltham, Massachusetts, USA) (range: ~ 40 and ~ 90 μg), and 400 μL of chilled 100 mM potassium phosphate buffer (pH 7.30) was added per 1 mg. After brief sonication on ice, a small volume was subjected to enzymatic analyses for LDH activity as performed for the homogenates (expressed as μmol/min/g dw).

Statistical analyses

All results are presented as mean ± standard deviation. Statistical comparisons were performed using either the Friedman repeated measures ANOVA with a Dunns *post hoc* test or Wilcoxon signed rank *t*-test for non-parametric paired data. Significance was set at $P<0.05$. Correlation coefficients were calculated using the two-tailed Pearson's correlation test. In some instances, the sample size was <18 due to the small biopsy size precluding the analysis of all parameters.

Results

HIIT protocol

The number of intervals completed per HIIT session varied during the initial 2 weeks, as athletes found it difficult to complete all six. However, after the second week, all athletes were able to complete the prescribed six intervals per day. HIIT total running time and total distance amounted to 155 ± 48 min and 51.8 ± 16.6 km, respectively. HIIT only replaced certain outdoor training sessions; those that would have been planned for that particular day, leaving the outdoor training affected as little as possible by the intervention. Of the total distance training performed per week (km) during the intervention period, HIIT amounted to ~ 17% of the total distance.

Physiological characteristics and maximum exercise tests

The physiological, training and performance variables of 10 athletes pre-screened three months before, before and after the HIIT intervention, are reported in Table 1a. Apart from a significant increase in PTS after HIIT ($P<0.01$), the following parameters remained the same: training volume, 10 km field test and maximum exercise test results within this period of time. This finding would therefore indicate that the HIIT had the most significant impact on the physiological and biochemical measurements. For all 18 athletes, body mass was not affected by the intervention (Table 1b). After HIIT, PTS increased by 5.2% ($P<0.001$), HR_{max} decreased with a concomitant increase in oxygen pulse. No other variable was affected by the intervention: RER_{max}, $\dot{V}O_{2\,max}$ or $\dot{V}_{E\,max}$ did not change significantly in response to HIIT. A correlation was observed between PTS and the 10 km field test time ($r=-0.86$, $P<0.001$), as well as between $\dot{V}O_{2\,max}$ and 10 km field test time ($r=-0.63$, $P<0.01$). Furthermore, the change in PTS was correlated with the HIIT per week ($r=0.65$, $P<0.01$).

Submaximum exercise tests

Table 2 and Fig. 1 report the submaximum exercise tests before and after the intervention. $\dot{V}O_2$ and RER

Table 1a. Training and maximal exercise data 3 months prior, before and after HIIT of 10 athletes

	Three months prior	Before HIIT	After HIIT
Outdoor training			
Training volume (km/week)	54 ± 18	53 ± 21	57 ± 27
10 km field test (min)	35.8 ± 2.4	35.7 ± 2.2	NP
Maximum exercise tests			
PTS (km/h)	21.1 ± 1.0	21.1 ± 0.8	22.2 ± 1.5*
$\dot{V}O_{2\,max}$ (mL/min/kg)	67 ± 4	67 ± 5	69 ± 3
HR_{max} (beats/min)	194 ± 8	190 ± 6	190 ± 4
RER_{max}	1.16 ± 0.03	1.15 ± 0.03	1.14 ± 0.04
Oxygen pulse (mL/beat)	22.7 ± 2.7	23.2 ± 2.7	23.7 ± 2.3
$\dot{V}_{Emax}$ (L/min)	156 ± 22	156 ± 18	162 ± 17

Different from 3 months prior and before HIIT $^*P<0.01$.

HR, heart rate; ITS, interval training speed; np, not performed; RER, respiratory exchange ratio; PRD_A, average preferred racing distance; PTS, peak treadmill speed; $\dot{V}O_2$, oxygen consumption; $\dot{V}_E$, minute ventilation; HIIT, high-intensity interval training.

Table 1b. Training and maximal exercise data before and after HIIT of all 18 athletes

	Before HIIT	After HIIT
Outdoor training		
Training volume (km/week)	56 ± 20	50 ± 26
10 km field test (min)	35.6 ± 2.8	NP
Maximum exercise tests		
PTS (km/h)	21.0 ± 0.8	22.1 ± 1.2**
$\dot{V}O_{2\,max}$ (mL/min/kg)	67 ± 5	68 ± 3
HR_{max} (beats/min)	190 ± 7	187 ± 7*
RER_{max}	1.16 ± 0.04	1.15 ± 0.04
Oxygen pulse (mL/beat)	21.3 ± 3.3	22.0 ± 3.3*
$\dot{V}_{Emax}$ (L/min)	144 ± 24	147 ± 24

Different from before HIIT, *$P<0.05$, **$P<0.001$.
HR, heart rate; ITS, interval training speed; NP, not performed; RER, respiratory exchange ratio; PRD_A, average preferred racing distance; PTS, peak treadmill speed; $\dot{V}O_2$, oxygen consumption; $\dot{V}_E$, minute ventilation; HIIT, high-intensity interval training.

Table 2. Submaximal exercise tests before and after HIIT ($n = 18$) with workload set as a percentage of the baseline PTS

	Before HIIT	After HIIT
	64% of PTS	
Treadmill speed (km/h)	13.4 ± 0.5	
$\dot{V}O_2$ (mL/min/kg)	50 ± 3	49 ± 3
HR (beats/min)	158 ± 11	146 ± 12**
RER	0.91 ± 0.03	0.90 ± 0.02
Oxygen pulse (mL/beat)	19.2 ± 3.3	20.3 ± 3.0*
	72% of PTS	
Treadmill speed (km/h)	15.1 ± 0.6	
$\dot{V}O_2$ (mL/min/kg)	55 ± 3	55 ± 3
HR (beats/min)	172 ± 11	161 ± 13**
RER	0.95 ± 0.03	0.94 ± 0.03
Oxygen pulse (mL/beat)	19.4 ± 3.2	20.6 ± 2.9*
	80% of PTS	
Treadmill speed (km/h)	16.8 ± 0.6	
$\dot{V}O_2$ (mL/min/kg)	60 ± 4	60 ± 3
HR (beats/min)	180 ± 11	172 ± 12**
RER	0.99 ± 0.04	0.97 ± 0.03
Oxygen pulse (mL/beat)	20.0 ± 3.8	21.0 ± 3.1*

Different from before HIIT, *$P<0.05$, **$P<0.001$.
HR, heart rate; RER, respiratory exchange ratio; $\dot{V}O_2$, oxygen consumption; PTS, peak treadmill speed; HIIT, high-intensity interval training.

did not change, but HR and oxygen pulse were lower and higher, respectively, at each workload after HIIT (both $P<0.01$). Plasma lactate was lower at 64% and 80% PTS, but not at 72% PTS (Fig. 1). The change in oxygen pulse did not correlate with the change in plasma lactate at any workload.

Muscle morphology and oxidative capacity

Fiber type (calculated from single fibers and MHC isoforms in homogenate samples) was unaffected by HIIT (Table 3). Conversely, type II fiber CSA tended to have decreased ($P = 0.06$) in response to HIIT, whereas capillaries per fiber remained the same. CS and 3-HAD activities were unaffected by HIIT.

LDH activities in homogenate and single fiber pools

LDH activities in homogenate and type IIa fibers increased significantly after HIIT (Table 3 and Fig. 2). The change in homogenate LDH activity did not correlate with ITS, but the relationship was significant between the change in type IIa LDH activity and ITS (Fig. 3).

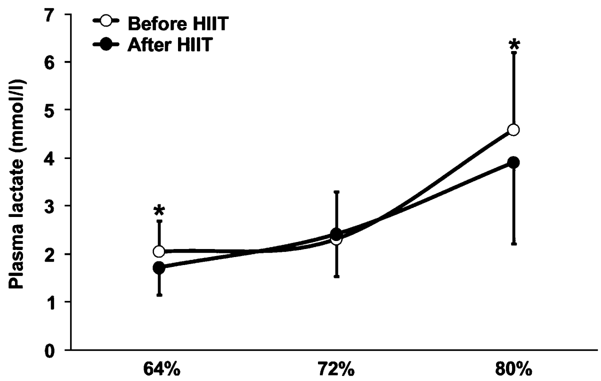

Fig. 1. Plasma lactate concentrations during the submaximal exercise tests before and after high-intensity interval training (HIIT). Values are means ± SD ($n = 18$). *Significantly different from before HIIT, $P < 0.05$.

Table 3. Skeletal muscle enzyme activities and morphology in homogenate muscle samples

	Before HIIT	After HIIT
Enzyme activities ($n = 18$)		
CS (μmol/min/g dw)	32.0 ± 8.7	30.8 ± 10.4
3-HAD (μmol/min/g dw)	20.7 ± 6.6	21.3 ± 7.7
LDH (μmol/min/g dw)	373 ± 76	394 ± 95*
Fiber type ($n = 14$)‡		
Type I (%)	47 ± 17	43 ± 12
Type I/IIa (%)	7 ± 4	7 ± 4
Type IIa (%)	33 ± 14	32 ± 15
Type IIa/IIx (%)	9 ± 8	14 ± 11
Type IIx (%)	2 ± 3	2 ± 5
Type Iax (%)	1 ± 1	1 ± 1
Type Ix (%)	1 ± 2	1 ± 1
Total hybrids (%)	18 ± 9	23 ± 10
MHC isoform content ($n = 18$)		
MHC I (%)	48 ± 11	49 ± 11
MHC IIa (%)	46 ± 11	45 ± 10
MHC IIx (%)	6 ± 9	6 ± 9
CSA and capillaries ($n = 10$)‡		
Type I fibers (μm^2)	4615 ± 1172	3927 ± 834
Type II fibers (μm^2)	5532 ± 1491	4603 ± 1157†
Capillaries/fiber	6.0 ± 0.9	5.6 ± 1.2

Different from before HIIT, *$P<0.05$, †$P = 0.06$.
‡Fiber type assessed from identification dissected single fibers and CSA and capillaries assessed histologically.
CS, citrate synthase; 3-HAD, 3-hydroxyacetyl coenzymeA dehydrogenase; CSA, cross-sectional area; LDH, lactate dehydrogenase; MHC, myosin heavy chain; HIIT, high-intensity interval training.

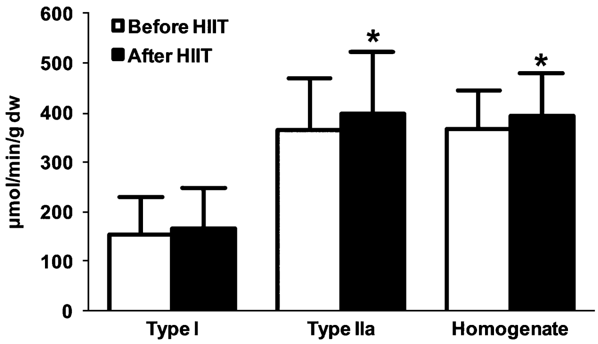

Fig. 2. Lactate dehydrogenase activity in pools of type I and type IIa fibres before and after high-intensity interval training (HIIT), in comparison to homogenate. Values are means ± SD ($n = 12$). *Significantly different from before HIIT, $P < 0.05$.

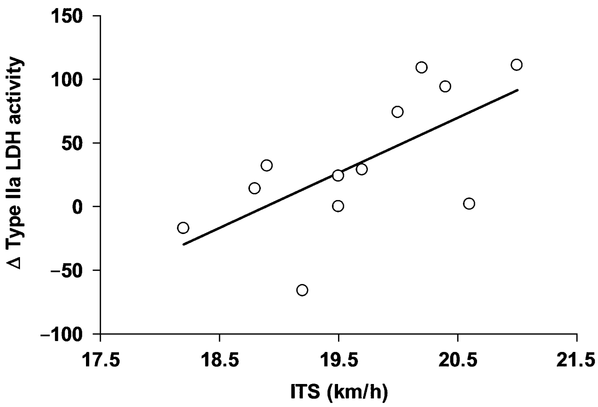

Fig. 3. Relationship between the change in lactate dehydrogenase (LDH) activity of type IIa fibres and ITS (Pearson's $r = 0.65$, $P < 0.05$, $n = 12$).

Discussion

The major findings were that 12 sessions of HIIT did not change $\dot{V}O_{2\,max}$, but increased PTS, and lowered HR and plasma lactate concentrations at submaximal exercise in already well-trained runners. Fiber type composition, CS and 3-HAD activities were unaffected, but LDH activity increased, especially in type IIa fibers. Further significant findings were related to the individualized HIIT; the total volume of interval training correlated with the improvement in PTS, but the change in type IIa fiber LDH activity correlated with ITS.

The HIIT protocol is well documented (Billat, 2001) and utilized with success to improve performance in well-trained endurance runners (Smith et al., 2003) and cyclists (Laursen et al., 2002). Because PTS is more closely related to racing performance than $\dot{V}O_{2\,max}$ (Noakes et al., 1990), the ITS of each athlete was set relative to individual PTS (94%), resulting in a higher training intensity and shorter interval duration than earlier HIIT studies on well-trained cyclists (Westgarth-Taylor et al., 1997; Stepto et al., 2001).

The athletes in this study were well trained and competitive. Three months before the intervention, 10 athletes were screened for performance and daily training activity (Table 1a). Although 3 months elapsed, both training status and performance (10 km field test and maximum exercise tests) remained the same and was the case for all 18 athletes (Table 1b). Additionally, the investigators instructed the coaches and athletes to abstain from altering their outdoor training programs and to follow their normal routine, as the HIIT would only make up a small portion of overall training. In subelite cyclists, replacing ~15% of their endurance training with HIIT increased laboratory cycling performance (Lindsay et al., 1996). The current study is in agreement, because runners replaced approximately 17% of their usual training with HIIT and improved their PTS.

It is not uncommon that no change in $\dot{V}O_{2\,max}$ is observed after HIIT in well-trained athletes, as was the case in the present study. For example, in both well-trained cyclists and middle-distance runners, performance improved, but $\dot{V}O_{2\,max}$ was unaltered after HIIT (Acevedo & Goldfarb, 1989; Laursen et al., 2002; Smith et al., 2003; Harber et al., 2004). It also appears that the number of sessions may not assist in changing $\dot{V}O_{2\,max}$. HIIT studies using mainly untrained or moderately trained recreational athletes usually show an improvement in performance accompanied by a significant increase in $\dot{V}O_{2\,max}$ (Kubukeli et al., 2002). This may indicate that, in well-trained athletes, the oxidative capacity is already high and any changes that may occur are too small to be detected in response to a training intervention.

PTS, rather than $\dot{V}O_{2\,max}$, correlated better with 10 km time, being in agreement with a previous study (Noakes et al., 1990). HR_{max} did decrease with 1.6% ($P = 0.04$) after HIIT, but was more significant when expressed as oxygen pulse (3.3%; $P = 0.02$) (Table 2). Agreeing with previous findings, it suggests more oxygen delivery per heart beat, likely from an increased stroke volume (Laffite et al., 2003). Lower HR at submaximal intensities after HIIT is in agreement with earlier studies and therefore one of the major physiological adaptations associated with HIIT (Acevedo & Goldfarb, 1989; Kubukeli et al., 2002).

No change in fiber type proportions (pure and hybrid) was observed after the intervention and was confirmed with the MHC isoform analyses in homogenate samples (Table 3). This is in agreement with previous HIIT studies using well-trained runners (Houston & Thomson, 1977) and skiers (Evertsen et al., 1999). Although HIIT was only performed for 6 weeks, it may be unlikely that significant fiber type changes will occur in well-trained runners. Supporting this argument is that 5 months of HIIT did not alter fiber type in well-trained cross country skiers (Evertsen et al., 1999).

In untrained subjects, an increase in muscle capillary to fiber ratio is observed after 4 weeks HIIT (Jensen et al., 2004). The number of capillaries around a fiber in the current study were high when compared with data from previous studies on endurance runners (Saltin et al., 1977). As no change occurred in capillaries per fiber as a result of HIIT (Table 3), the well-trained runners may already have an adapted capillary supply allowing for adequate blood flow to the working muscle.

The CSA of types I and IIa fibers increase in response to high-intensity strength training in untrained subjects, decrease with marathon training (Trappe et al., 2006), but only in type IIa fibers in subjects performing a combination of strength and endurance training (Putman et al., 2004). Very limited data are available on CSA of well-trained runners subjected to HIIT. One study found that HIIT reduced type I fiber diameter in competitive cross country runners without affecting the contractile properties (such as power) thereof (Harber et al., 2004). Only type II fibers showed a tendency to have decreased after HIIT (Table 3). It is still unclear why the CSA of fibers decrease as a result of HIIT, but may be important in overall muscle function, thus warranting a future investigation. One positive effect may be a faster release of lactate from the fibers and a faster uptake of oxygen from blood.

It is well known that subjecting the untrained to endurance training significantly increases CS and 3-HAD activities (Kubukeli et al., 2002). Because of the high activities of these enzymes in already well-trained runners, it was not an unexpected finding of no change in the muscles after HIIT, concurring with previous results of HIIT in well-trained cyclists (Weston et al., 1997).

Interestingly, LDH activity increased in homogenate muscle, but specifically in type IIa fiber pools (9.3%), suggesting that more type IIa fibers were recruited during HIIT, therefore increasing the demand for rapid ATP production. This is in contrast to what Houston and Thomson (1977) found where 6 weeks of HIIT elicited no change in LDH activity. However, it has been shown that LDH activity can increase with an improvement in performance as a result of intense anaerobic exercise, but may be dependable on the training intervention itself (Linossier et al., 1997). Yet, the exact mechanism is still under investigation (see the lactate shuttle hypothesis below).

Although plasma lactate changes in response to HIIT were small, they were evident at 64% and 80% PTS, which is in agreement with Acevedo and Goldfarb (1989). No difference at 72% PTS may reflect individual differences between athletes in lactate threshold. Nonetheless, the decreased plasma lactate concentrations correspond to $\sim 70\%$ and $\sim 90\%$ $\dot{V}O_{2\,max}$. The changes at these two markedly different exercise intensities may be a result of different mechanisms. For example, endurance training in untrained subjects decreased lactate appearance at lower intensities, but increased metabolic clearance at low and high intensities (MacRae et al., 1992). The complexity of lactate metabolism is dependent on factors such as lactate transporters, lactate kinetics and the proposed lactate shuttle systems (Brooks, 2000), all of which may be responsive to training albeit with responses that differ depending on the specific training status before intervention and the design of the intervention itself (Kubukeli et al., 2002).

The lactate shuttle system is based on the hypothesis that lactate, produced in muscle fibers, may be reconverted to pyruvate, either within the same fiber or by adjacent fibers (Brooks, 2000). This pyruvate may then enter the tricarboxylic acid cycle to be metabolized. Although some controversy exists (Rasmussen & Rasmussen, 2002), the structural evidence for a mitochondrial lactate oxidation complex with LDH on the outer membrane lends significant support to the interpretation of an intracellular lactate "shuttle" leading to lactate oxidation (Hashimoto & Brooks, 2008). The positive relationship between the changes in LDH activity in pools of type IIa single fibers and ITS (Fig. 3) suggests that LDH may play an important role in maintaining metabolic homeostasis despite high glycolytic flux during high-intensity exercise. The absolute speed, at which interval training is performed, appears to be an important determinant of the extent of LDH activity adaptation.

A recent study suggested that regular endurance training at a high intensity promotes the enhancement of maximal mitochondrial capacities to oxidize carbohydrate rather than fatty acids (Daussin et al., 2008). This is in agreement with our results on LDH and 3-HAD activity, suggesting that further adaptations in well-trained runners in response to HIIT are likely to occur in lactate metabolism rather than in fat oxidation. The capacity for fat and carbohydrate oxidation is already high in both types I and II fibers of well-trained athletes as shown by high 3-HAD and CS activities (Essén-Gustavsson & Henriksson, 1984). The high oxidative capacity in type II fibers is one reason explaining the ability of well-trained runners to perform at a high % of $\dot{V}O_{2\,max}$ or PTS before blood lactate markedly increases, and is supported by the relatively low blood lactate levels seen at 80% PTS (Fig. 1).

The amount of HIIT performed per week was related to the improvement in PTS, indicating that the total distance performed at high speed may be of importance for some adaptations that impact on performance. Other adaptations are observed in muscle after training at high speeds. In trained athletes, repeated intense training resulted in an improved buffering capacity (Weston et al., 1997) and alterations are observed in the Na^+–K^+ pump (Iaia et al., 2008). Furthermore, recent investigations focused on GLUT4 and enzymes involved in glycogen metabolism, also emphasize the importance of investigating adaptations in a fiber-specific manner (Daugaard & Richter, 2004).

Recent reports suggest that the success of South African black runners may be attributed to their higher reported training speeds compared with their white counterparts (Coetzer et al., 1993; Kohn et al., 2007a). In performance-matched and training volume-matched subjects, Kohn et al. (2007a) found higher LDH activity in both muscle homogenate and type IIa fiber pools, alongside lower plasma lactate concentrations at submaximal exercise intensity in the black runners. The findings of the present longitudinal study suggest that differences in training intensity may have been one of the factors underlying those findings (Kohn et al., 2007a).

In conclusion, 6 weeks of HIIT elicited significant enhancements in performance and increased LDH activity in muscle and especially in type IIa fibers. This study is the first to suggest that absolute interval speed and cumulative interval duration may be associated with different adaptations to HIIT.

Perspectives

For future studies to verify these results, they should be designed around the absolute running speed of the HIIT bouts (or other workload for other exercise models), rather than the relative workload. In well-trained endurance runners, adaptations to HIIT are not to further increase oxidative enzyme activities as is the case in less-trained individuals, but rather to enhance the activity of the enzyme central to intracellular lactate metabolism (LDH). This was apparent only in fast twitch fibers and more apparent at a high interval speed. Therefore, whether or not acute exercise or longer term-training studies are undertaken, the analysis should include fiber type-specific responses.

Key words: peak treadmill speed, plasma lactate, oxygen pulse, training intervention, citrate synthase, single fibers.

Acknowledgements

The authors would like to thank the athletes for their participation and Dr. J. Brink for performing the muscle biopsies. Thanks are extended to Kristina Karlström (Swedish Agricultural University) and Karen van Tubbergh (Stellenbosch University) for assisting with the histological preparations. Gabriel Eksteen, Robyn Bowen, Michael Flynn and Jo-Anne du Toit are thanked for assistance during exercise testing. This study was supported by the Harry Crossley Foundation, Swedish Institute Scholarship to Tertius A. Kohn (2000) and a bilateral research grant from the National Research Foundation of South Africa and Swedish International Development Cooperation Agency.

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